Final

Group A & B

(a) Describe absolute value and argument of a complex number.

Absolute value/modulus: If $z = x+iy$ be a complex number, then modulus or absolute value of z is denoted by $|z|$ or r and is defined by $|z| = r = \sqrt{x^2+y^2}$.

Arguement/Amplitude: If $z = x+iy$ be a complex number, then the arguement or amplitude of z is denoted by $\arg(z)$ or $\operatorname{amp}(z)$ or θ and is defined by $\arg(z) = \theta = \tan^{-1}\frac{y}{x}$.

Two complex number z_1, z_2, Prove $|z_1 z_2| = |z_1||z_2|$

Solution:

Let in polar form $z_1 = r_1 e^{i\theta_1}$ ~~arg(z)~~ $\therefore |z_1| = r_1$

$z_2 = r_2 e^{i\theta_2}$ $\therefore |z_2| = r_2$

Now, $z_1 z_2 = r_1 e^{i\theta_1} . r_2 e^{i\theta_2} \Rightarrow r_1 r_2 e^{i\theta_1} . e^{i\theta_2}$

$= r_1 r_2 e^{i(\theta_1+\theta_2)}$

$\Rightarrow |z_1 z_2| = r_1 r_2$

$\therefore |z_1 z_2| = |z_1||z_2|$ proved.

(b) Two complex numbers z_1 and z_2, prove that

$$\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$$

Solution:

Let in polar form,

$$z_1 = r_1 e^{i\theta_1} \quad \therefore \arg(z_1) = \theta_1$$

$$z_2 = r_2 e^{i\theta_2} \quad \therefore \arg(z_2) = \theta_2$$

Now,
$$\frac{z_1}{z_2} = \frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}}$$

$$= \frac{r_1}{r_2} e^{i\theta_1 - i\theta_2}$$

$$= \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$$

$$\Rightarrow \arg\left(\frac{z_1}{z_2}\right) = \theta_1 - \theta_2$$

$$\therefore \arg\left(\frac{z_1}{z_2}\right) = \cancel{(\arg z_1)} \arg(z_1) - \arg(z_2)$$

(c) Given

$z_1 = 3+4i$

$z_2 = 4-3i$

$$L.H.S = |z_1+z_2|^2 + |z_1-z_2|^2$$
$$= |3+4i+4-3i|^2 + |3+4i-4+3i|^2$$
$$= |7+i|^2 + |-1+7i|^2$$
$$= \left(\sqrt{7^2+1^2}\right)^2 + \left(\sqrt{(-1)^2+7^2}\right)^2$$
$$= 49+1+1+49$$
$$= 100$$

$$R.H.S = 2|z_1|^2 + 2|z_2|^2$$
$$= 2|3+4i|^2 + 2|4-3i|^2$$
$$= 2\left(\sqrt{3^2+4^2}\right)^2 + 2\left(\sqrt{4^2+(-3)^2}\right)^2$$
$$= 2(9+16) + 2(16+9)$$
$$= 50+50$$
$$= 100$$

$\therefore$ ~~L.H.S = R.H.S verif~~

$$|z_1+z_2|^2 + |z_1-z_2|^2 = 2|z_1|^2 + 2|z_2|^2$$

(verified)

(d)

$z = x + iy$

Given, $3|z-1| = 2|z-2|$

$\Rightarrow 3|x+iy-1| = 2|x+iy-2|$

$\Rightarrow 3|(x-1)+iy| = 2|(x-2)+iy|$

$\Rightarrow 3\sqrt{(x-1)^2+y^2} = 2\sqrt{(x-2)^2+y^2}$

$\Rightarrow 9\{(x-1)^2+y^2\} = 4\{(x-2)^2+y^2\}$

$\Rightarrow 9\{x^2-2x+1+y^2\} = 4\{x^2-4x+4+y^2\}$

$\Rightarrow 9x^2-18x+9+9y^2-4x^2+16x-16-4y^2=0$

$\Rightarrow 5x^2+5y^2-2x-7=0$

$\therefore 5x^2+5y^2-2x-7=0$

This represents a circle.

Explain analytic function. Show that the function $f(z) = e^x(\cos y + i\sin y)$ is analytic and find its derivative

Answer

Analytic Function

A Function $f(z)$ is said to be analytic at a point if it is differentiable at that point and in a neighbourhood of that point.

If $f(z) = u(x,y) + iv(x,y)$

Then it is analytic if :

$$u_x = v_y$$
$$u_y = -v_x$$

(Cauchy-Riemann Equations)

and the partial derivatives are continuous.

Given Function

$$f(z) = e^x(\cos y + i\sin y)$$

Separate Real and Imaginary parts

$$f(z) = e^x\cos y + ie^x\sin y$$

So,
$$u(x,y) = e^x\cos y$$
$$v(x,y) = e^x\sin y$$

Partial Derivatives

$$u_x = \frac{\partial}{\partial x}(e^x \cos y)$$

$$= e^x \cos y$$

$$u_y = \frac{\partial}{\partial y}(e^x \cos y)$$

$$= -e^x \sin y$$

$$v_x = \frac{\partial}{\partial x}(e^x \sin y)$$

$$= e^x \sin y$$

$$v_y = \frac{\partial}{\partial y}(e^x \sin y)$$

$$= e^x \cos y$$

Verify C-R equations

$$U_x = e^x \cos y$$

$$V_y = e^x \cos y$$

$$\therefore U_x = V_y$$

$$U_y = -e^x \sin y$$

$$-V_x = -e^x \sin y$$

$$\therefore U_y = -V_x.$$

Since C-R equations are satisfied and derivatives are continuous, $f(z)$ is analytic everywhere.

We know, $f'(z) = u_x + iv_x$

$$= e^x \cos y + ie^x \sin y$$

$$\therefore f'(z) = e^x(\cos y + i\sin y)$$

<u>F</u>

Discuss harmonic and conjugate harmonic. Show that the function $u = \sin x \cosh y$ is harmonic. Determine its harmonic conjugate v, also construct the corresponding analytic function $f(z)$.

<u>Answer</u>

Harmonic Function

A real-valued function $u(x,y)$ is called harmonic if it satisfies Laplace's equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Conjugate Harmonic Function

If $u(x,y)$ and $v(x,y)$ satisfy the Cauchy-Riemann equations:

$$u_x = v_y$$
$$u_y = -v_x$$

Then v is called the harmonic conjugate of u, $f(z) = u + iv$ is an analytic function.

Given, $u = \sin x \cosh y$

Find the Partial Derivatives

$$u_x = \cos x \cosh y$$
$$u_{xx} = -\sin x \cosh y$$

Find the derivative of y:

$$u_y = \sin x \sinh y$$
$$u_{yy} = \sin x \cosh y$$

Check Laplace equation

$$u_{xx} + u_{yy} = (-\sin x \cosh y) + (\sin x \cosh y)$$
$$= 0.$$

Therefore, u is harmonic.

Find harmonic conjugate v using C-R equations:

$$u_x = v_y$$
$$u_y = -v_x$$

We have: $u_x = \cos x \cosh y$

So, $u_y = \cos x \cosh y$

Integrate w.r.t y:

$$v = \cos x \sinh y + \phi(x)$$

Where $\phi(x)$ is function of (x).

Now use second C-R equation :

$$u_y = \sin x \sinh y$$
$$u_y = -V_x$$

Find u_x :

$$u_x = -\sin x \sinh y + \phi'(x)$$
$$-V_x = \sin x \sinh y - \phi'(x)$$
$$u_y = \sin x \sinh y$$

Therefore,

$$\sin x \sinh y = \sin x \sinh y - \phi'(x)$$

so, $\phi'(x) = 0$

$$\phi'(x) = c$$

Harmonic Conjugate $u = \cos x \sinh y$

Construct Analytic Function

$$f(z) = u + iv$$
$$= \sin x \cosh y + i \cos x \sinh y$$

But we know identity :

$$\sin(x+iy) = \sin x \cosh y + i \cos x \sinh y$$

so, $f(z) = \sin z$

Now,

$u = \sin x \cosh y$ is harmonic

Harmonic conjugate :

$$u = \cos x \sinh y$$

Corresponding analytic function :

$$f(z) = \sin z$$

Q

Show that the real and imaginary part of a complex function $f(z) = u + iv$ is harmonic.

Answer

Let, $f(z) = u(x,y) + iv(x,y)$ be analytic. Then u and v satisfy the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial u}{\partial x}.$$

The laplacian of u is :

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right)$$

$$= \frac{\partial^2 v}{\partial x \partial y} - \frac{\partial^2 v}{\partial y \partial x}$$

Since the order of differentiation doesnot matter, the righ side becomes 0. Thus :

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \text{ so } u \text{ is hamonic}$$

Showing the Imaginary part (v) is Harmonic

Differentiate the first equation with respect to y:

$$\frac{\partial}{\partial y}\left(\frac{\partial v}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right)$$

$$\Rightarrow \frac{\partial^2 v}{\partial y^2} = \frac{\partial^2 u}{\partial y \partial x}$$

Differentiate the second equation with respect to x:

$$\frac{\partial}{\partial x}\left(\frac{\partial v}{\partial x}\right) = \frac{\partial}{\partial x}\left(-\frac{\partial u}{\partial y}\right)$$

$$\Rightarrow \frac{\partial^2 v}{\partial x^2} = -\frac{\partial^2 u}{\partial x \partial y}$$

Adding them together:

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = -\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y \partial x} = 0$$

This proves v is also harmonic.

Group B

(h) Explain Fourier series. Discuss the application of Fourier series.

Explaination of Fourier Series

A Fourier Series is a way to represent a periodic function as an infinite sum of simpler sine and cosine waves. It was developed by Jean-Baptiste Joseph Fourier, who discovered that almost any periodic signal can be broken down into these basic buiding blocks.

Mathematical Defination :

If we have a function $f(x)$ that repeats itself over a period of $2L$, we can write it as:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)\right)$$

(i) a_0, a_n, b_n: These are called the Fourier coefficient They tell us "how much" of each sine and cosine wave is needed to rebuild the original function.

(ii) $\frac{a_0}{2}$: This represents the average value (or the "DC offset") of the function over one period.

(iii) Harmonics : The terms where $n > 1$ represent higher frequencies that add detail and shape to the wave. In simple terms, just like a musical chord is made up of several individual notes, a complex wave is made up a Fourier Series of simple sine and cosine waves.

Application of fourier series

Fourier series are incredibly useful in science and engineering because they allow us to analyze complex signals by looking at their individual frequency components.

1. Signal Processing: In telecommunications, Fourier series are used to analyze and transmit signals. For example, your phone converts your voice into frequency components to send it over the air and then reconstucts it on the other side.

2. Audio and Image Compression:
Formats like MP3 and JPEG use Fourier-related transforms to remove frequencies that the human ear or eye cannot perceive, making the files much smaller without losing noticeable quality

3. Heat Transfer :

Fourier originally developed this series to solve the "Heat equation". It helps scientists understand how temperature changes and spreads through a solid object over time.

4. Acoustics (Music) :

It is used to analyze the 'timbre' or quality of a sound. It explains why a violin and a piano sound different even when playing the same note, they have the same fundamental frequency but different "harmonies".

5. Vibration Analysic :

Engineers use Fourier series to study the vibrations in building, bridge and engines. By identifying the frequencies that cause the most vibration, they can design structures that are safer and won't break.

6. Quantum Physics :

In quantum mechanics, particles are often described as waves. Fourier series help in solving the Schrodinger equation to find the energy levels of atoms.

① Estabilish the fourier series for even function and odd function.

Answer:

To establish the Fourier series for even and odd function, we first recall the general Fourier series for a function $f(x)$ with period $2L$ on the interval $[-L, L]$:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right)$$

Where the coefficients are given by:

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\,dx, \quad a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx,$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}\,dx$$

1. Fourier Series for an Even Function

A Function is even if $f(-x) = f(x)$. Its graph is symmetrical about the y-axis (like $\cos(x)$ or x^2).

- Properties of Integrals: For an even function, ~~$f(x)$~~ the integral over $[-L, L]$ is twice the integral over $[0, L]$.
- The sine Term (b_n): The product of an even function

$f(x)$ and odd function $\sin\frac{n\pi x}{L}$ is an odd function. The integral of an odd function from $-L$ to L is always zero.

$$b_n = \frac{1}{L}\int_{-L}^{L} (\text{even}) \times (\text{odd})\, dx = 0.$$

The Fourier Consine series:

Since $b_n = 0$, the series contains only cosine terms:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{L}.$$

Where:

$$a_0 = \frac{2}{L}\int_0^L f(x)\, dx, \quad a_n = \frac{2}{L}\int_0^L f(x)\cos\frac{n\pi x}{L}\, dx.$$

2. Fourier Series for an Odd function

A function is odd if $f(-x) = -f(x)$. Its graph is symmetrical about the origin (like $\sin(x)$ or x^3).

The Constant and Cosine Terms (a_0, a_n): The product of an odd function $f(x)$ and an even function $\cos\frac{n\pi x}{L}$ is an odd function. Thus, the integrals over $[-L, L]$ are zero.

$$a_0 = \frac{1}{L}\int_{-L}^{L} (\text{Odd})\, dx = 0$$

$$a_n = \frac{1}{L}\int_{-L}^{L} (\text{odd}) \times (\text{even})\, dx = 0$$

The sing term (b_n): The product of two odd functions ($f(x)$ and $\sin\frac{n\pi x}{L}$) is an even function.

$$b_n = \frac{1}{L}\int_{-L}^{L} (\text{Odd}) \times (\text{Odd})\, dx = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}\, dx$$

The Fourier Sine series:

Since $a_0 = 0$ and $a_n = 0$, the series contains only sine terms:

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}$$

Where,

$$b_n = \frac{2}{L}\int_0^L f(x) \sin \frac{n\pi x}{L}\, dx$$

Group c

The **Laplace Transform** is a powerful mathematical tool used to turn difficult differential equations (calculus) into simpler algebraic equations (basic math). This makes it essential in various fields of science and engineering.

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1. Solving Differential Equations

The most direct application is solving linear ordinary differential equations (ODEs), especially those with **initial conditions**.

- Instead of dealing with derivatives, the Laplace transform converts them into powers of a new variable .
- After solving the simpler equation, we use the **Inverse Laplace Transform** to get the final answer.

2. Control Systems Engineering

Laplace transforms are the language of control systems.

- **Transfer Functions:** Engineers use them to create a "Transfer Function" that describes how a system (like a car's cruise control or a heating system) responds to an input.
- **Stability Analysis:** By looking at the "poles" (specific values of), engineers can tell if a system will stay steady or spiral out of control.

3. Electrical Circuit Analysis

It is used to analyze complex circuits containing resistors (), capacitors (), and inductors ().

- It simplifies the study of **transient response** (what happens the moment you flip a switch) and **steady-state response**.
- It turns complex integrodifferential equations of a circuit into simple Ohm's Law-like algebraic equations ().

4. Mechanical Engineering (Vibrations)

In mechanical systems, Laplace transforms help model:

- **Mass-Spring-Damper Systems:** Predicting how a car's suspension will react to a bump in the road.
- **Structural Resonance:** Ensuring that buildings or bridges don't vibrate at frequencies that could cause them to collapse.

5. Signal Processing

While Fourier transforms are used for steady signals, Laplace transforms are used for signals that might grow or decay over time.

- It helps in designing **filters** that can remove noise from a signal or enhance specific parts of a communication.

6. Process Control (Chemical Engineering)

In chemical plants, Laplace transforms are used to model the behavior of:

- Temperature in a chemical reactor.
- Fluid levels in tanks.
- Pressure changes in a piping system